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ORIGINAL RESEARCH

Estimation of Endurance During Elbow Flexion With Myoelectric Orthotic Assistance

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Abstract

Objectives: This study evaluated the impact of a powered myoelectric elbow orthosis (PMEO) on fatigue resistance and maximum endurance time (MET) of the gracilis muscle used in free functioning muscle transfer (FFMT) for elbow reconstruction. The PMEO was developed to support elbow flexion by mechanically locking the joint in place, reducing muscular demand during sustained tasks.

Design: Experimental clinical study.

Setting: Academic medical center.

Participants: Seven participants (N=7) with brachial plexus injury and FFMT were enrolled and divided into more and less impaired groups based on manual muscle test scores.

Interventions: Participants performed repeated isometric elbow flexion trials with and without the PMEO while surface electromyography (EMG) and kinematic data were recorded.

Main Outcome Measures: Spectro-temporal EMG metrics—median power frequency and integrated EMG—were analyzed, and endurance was estimated using a muscle-specific MET model.

Results: Four of the 7 participants, primarily in the more impaired group, demonstrated a 4-minute improvement in MET and more favorable fatigue metrics with the PMEO. Median power frequency increased, and integrated EMG decreased in these individuals, indicating delayed fatigue onset. The PMEO also enabled task completion in 3 participants who were otherwise unable to flex their elbow unaided.

Conclusions: These results suggest that the PMEO can enhance functional endurance in individuals with brachial plexus injury and FFMT. Notably, this study provided one of the first applications of spectro-temporal metrics and MET to evaluate fatigue resistance in an FFMT, filling a gap in existing literature.

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The brachial plexus is a network of peripheral nerves that controls motor and sensory function in the upper extremities and is commonly injured through high-energy trauma to the neck or shoulder, resulting in brachial plexus injury (BPI).[1] Surgical reconstruction techniques, such as nerve grafts, nerve transfers, and free functioning muscle transfers (FFMT), are used to restore function, with the gracilis muscle from the thigh often transferred to re-establish elbow flexion.[2] Despite these interventions, outcomes are frequently limited. Up to 40% of patients do not regain sufficient strength in the elbow flexor to overcome gravity after typical recovery periods,[3] highlighting the need for adjunctive strategies to enhance functional recovery.

To address the limitations of biological recovery, technological solutions such as powered orthoses have emerged. A powered myoelectric elbow orthosis (PMEO) has been developed to augment elbow function during activities of daily living (ADLs).[4] The PMEO is designed to hold the elbow in a flexed position using a mechanical brake, thereby reducing the need for continuous muscle activation during tasks that require sustained flexion, such as holding a phone or lifting an object to the mouth.[5] By offloading the demand on the elbow flexor, the device may reduce fatigue and extend the functional endurance of the muscle.

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Muscle endurance is typically assessed using maximum endurance time (MET), which quantifies the longest duration a muscle can sustain a given force output before failure.[6] Estimation of endurance time has been performed using various methods. Mehra et al[7] used a support vector regression−based algorithm to estimate endurance time using features derived from EMG frequency spectrum (similar to median frequency). Scano et al[8] used the median frequency of the EMG along with the blood's oxygen saturation and deoxy-hemoglobin metrics as descriptors of muscle fatigue and endurance. Hagberg described the relationship between EMG and endurance times at various levels of maximum voluntary contraction (MVC).[9] Other spectral and temporal parameters, such as median and mean power frequency, root mean square value, and mean absolute value, have also been used to estimate the muscle's fatigue resistance.[10-12]

In the context of upper limb rehabilitation, improving MET can directly impact a patient's ability to perform daily tasks without fatigue.[13-15] The use of powered orthotics for the upper extremities has also shown functional gains in ADLs in patients with a paralyzed arm.[16,17] However, there are limited data on how assistive technologies such as the PMEO affect the endurance properties of transferred muscles, such as the gracilis.

The objective of this pilot study was to evaluate the effect of the PMEO on the endurance and fatigue resistance of the gracilis muscle in individuals who underwent FFMT for elbow flexion reconstruction. Notably, this study provided one of the first applications of spectro-temporal metrics and MET to evaluate fatigue resistance in an FFMT, which fills a research gap in existing literature.

Methods

The PMEO

The PMEO is designed with a low-profile, lightweight form factor intended to be worn under clothing. The total weight of the orthotic components on the arm is ∼0.54 kg (±0.03 kg).[4] To further reduce distal weight and fatigue, the battery and controller, along with the mechanical brake, are housed in a separate backpack. The orthosis is engineered and tested to withstand a minimum elbow torque of 15 Nm.[5] In functional testing, it enabled highly impaired patients (manual muscle testing [MMT]<3) to lift and carry an average of 1.14±0.57 kg, whereas these individuals could lift no weight independently.[4] These participants also experienced a significant improvement in elbow flexion of 14±23° (P=.019), enabling them to flex their elbow beyond 90°.[4] The device uses surface electromyography sensors positioned on the elbow flexor to trigger movement. A custom software algorithm uses a "double pulse" routine to release the brake, allowing the arm to extend under gravity.[5]

List of abbreviations:

ADL	**activities of daily living**
BPI	**brachial plexus injury**
EMG	**electromyograph (muscle activity)**
$EMG_{norm_{GR}}$	**average normalized electromyograph for the gracilis**
$\Delta MET_{avg_{GR}}$	**difference in average maximum endurance time for the gracilis with the device compared with without**
$EMG_{norm_{KE}}$	**average normalized electromyograph for the knee extensor**
FFMT	**free functioning muscle transfer**
f_{MVC}	**percent force generated by the muscle**
$f_{MVC_{GR}}$	**percent force generated by the gracilis**
$f_{MVC_{KE}}$	**percent force generated by the knee extensor**
$iEMG_{GR}$	**integrated electromyograph for the gracilis**
MET	**maximum endurance time**
MET_{avg}	**average maximum endurance time**
$MET_{avg_{GR}}$	**average maximum endurance time for the gracilis**
MMT	**manual muscle testing**
MPF	**median power frequency**
MPF_{GR}	**median power frequency for the gracilis**
MVC	**maximum voluntary contraction**
PMEO	**powered myoelectric elbow orthosis**
τ_{GR}	**specific tension for the gracilis**
τ_{KE}	**specific tension for the knee extensor**

Participants

The data for this study were collected during the performance of a companion study.[4] Patients from the Brachial Plexus Injury Clinic at Mayo Clinic, Rochester, MN, were screened for eligibility to participate in this study. Eligible participants were aged between 18 and 65 years and had sustained a traumatic BPI resulting in loss of elbow flexion that required an FFMT surgery. Additional inclusion criteria included a functional passive range of motion in the affected upper limb and the cognitive ability to follow simple instructions. Exclusion criteria included closed head injuries that impaired command-following, musculoskeletal injuries that prevented orthosis use, nonfunctional passive range of motion, neuropathic pain that contraindicated powered exoskeleton use, and any unwillingness or inability to comply with study procedures. The recruited participants were divided into a more impaired and a less impaired group based on their unaided ability to flex their elbow against gravity.

Participants meeting these criteria, regardless of sex, race, or ethnicity, were contacted for recruitment. All participants were informed of the study procedures and provided written informed consent under the approval of the Mayo Clinic Institutional Review Board (IRB no. 20-006849). A board-certified orthotist from Limb Lab (Rochester, MN) designed and fitted each participant with custom orthotic components to interface the PMEO with the affected arm (fig 1A).

Data collection

The study was conducted in the Motion Analysis Laboratory at Mayo Clinic, Rochester, MN. Demographic and clinical data were collected for each participant, including age, height, weight, time since surgery, side of orthosis use, nerve used for transfer, and MMT scores assessed using the British Medical Research Council scale. An MMT grade 3 indicated the ability of the elbow to flex through its complete range of motion against gravity. The more impaired group had an MMT grade of <3, whereas the less impaired group had an MMT grade of ≥3.

A calibration session was performed to establish each participant's signal activation threshold for the PMEO and to configure control algorithm parameters, including the double pulse routine used to disengage the brake mechanism (fig 1B).[5,18] During this training session, participants were also familiarized with the device's control strategy[5] and overall operation. A surface electromyography (EMG) sensor with a built-in amplifier (EMG500; Motion Lab Systems, Inc.)[a] was positioned over the belly of the elbow flexor muscle (gracilis after the FFMT procedure) using transparent adhesive film (3M Tegaderm 1624W;

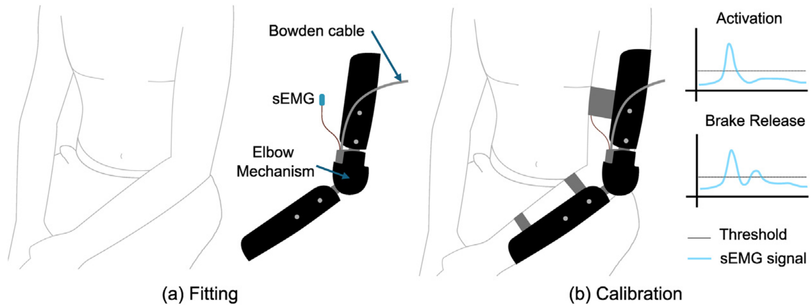

Fig 1 An illustration describing the initial setup process for the PMEO. (A) The PMEO was fit for each participant by a certified orthotist. The parts attaching the PMEO to the upper and lower arm (black) were vacuum formed using a thermoplastic. (B) A calibration procedure was performed to find the activation threshold and the brake release duration for each participant.[4] Abbreviation: SEMG, surface electromyography sensor.

3M).[b] An athletic prewrap (Tape Underwrap SKU: 214546; Cramer Sport Medicine)[c] was applied to secure the sensor in place. The activation threshold for the orthosis was set above the resting EMG level (quiescent) observed with the arm extended. The participants were also asked to perform an MVC.

Kinematic data were collected using a 12-camera motion capture system (Raptor-12, Motion Analysis Corporation).[d] Retro-reflective markers were placed on key anatomical landmarks to model the trunk, bilateral scapulas, upper arms, forearms, and hands,[19] allowing accurate measurement of elbow angles throughout testing.

To assess the amount of fatigue in the elbow flexor, the participants were asked to flex their elbow and hold it (similar to holding a grocery basket/bag or a purse; fig 2) for as long as they felt comfortable while holding a weighted basket. The basket was incrementally loaded with water bottles (489 g each) until the participant reached their functional limit or reported discomfort.[4] This maximum weight served as a representation of the fatigue experienced by the participant's elbow flexors at peak strength. The participants were asked to repeat the task 3 times with adequate time in between to relax/replenish their elbow flexor. The participant's success or failure to perform the task in each trial was recorded.

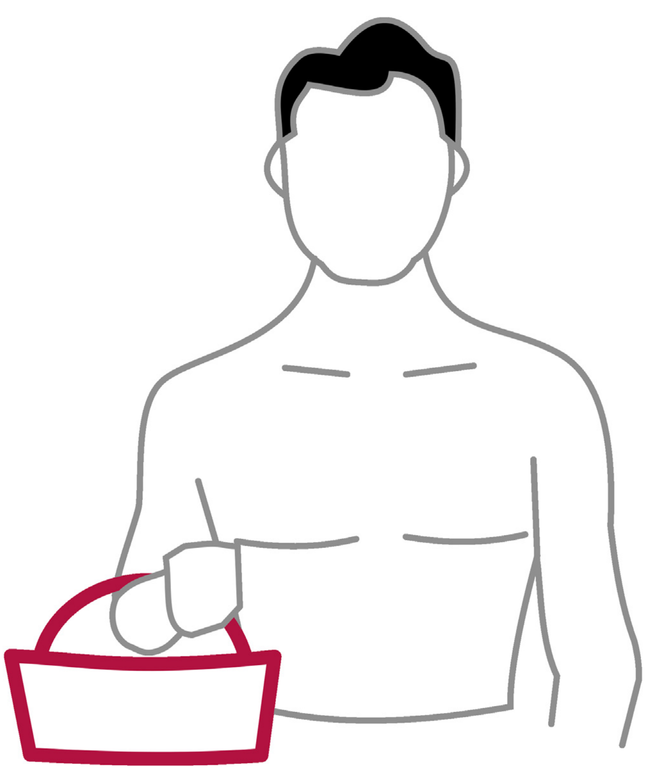

Fig 2 Activity performed by the participants to emulate holding a grocery basket/bag or a purse, aka any activity that requires a sustained elbow flexion.[4]

Data processing

Kinematics

Commercial biomechanical modeling software (Visual 3D; C-Motion, Inc)[e] was used to process 3-dimensional marker trajectories and construct segment coordinate systems for calculating elbow kinematics. Marker data were smoothed using a generalized cross-validatory spline filter within Visual 3D to reduce noise and improve signal quality.[20] Segment definitions for the forearm, upper arm, and trunk, as well as joint angle conventions for elbow and humerothoracic kinematics, followed the standards set by the International Society of Biomechanics.[21] Isometric elbow flexion was defined as maintenance of the elbow angle within $\pm 5^{\circ}$. Start and end instances of isometric elbow flexion were labeled for the kinematic elbow flexion data, and the time stamps for the instances were noted.

EMG spectro-temporal metrics

The EMG data were processed as per the recommendations of the International Society of Electrophysiology and Kinesiology.[22] The raw EMG data were bandpass filtered (fourth-order Infinite Impulse Response (IIR) Butterworth filter with bandpass filters set at 20 Hz and 450 Hz) and full-wave rectified. A low-pass filter (fourth-order IIR Butterworth filter with low-pass cut-off at 4 Hz) was used to obtain an envelope of muscle activation. The envelope was then normalized using the quiescent and MVC data. Using the time stamps for the start and end instances for isometric elbow flexion, average normalized EMG data ($EMG_{norm_{GR}}$) were calculated. The $EMG_{norm_{GR}}$ for trials where the participants were unable to perform the task were not included in the analysis. Median power frequency (MPF_{GR}) and integrated EMG ($iEMG_{GR}$) were calculated for the duration of the isometric elbow flexion. The participants who could not perform the tasks were assumed to have a low MPF_{GR} and high $iEMG_{GR}$ during analysis.

Maximum endurance time

Ahrache et al[6] reviewed multiple studies and compiled an equation for the average amount of MET per fraction of MVC force for the upper extremity

$$MET_{avg} = 705.5 \times f_{MVC}^{-1.601} \quad (1)$$

where MET_{avg} is the average MET in minutes and f_{MVC} is the force generated by the muscle as a percentage of the MVC. Kuroda et al[23] described the relationship between $f_{MVC_{KE}}$ and normalized muscle activity for knee extension $(EMG_{norm_{KE}})$

$$EMG_{norm_{KE}} = \alpha_{KE} \times f_{MVC_{KE}}/100 \quad (2)$$

Assuming the constant (α_{KE}) is related to the physiological constants of the muscles (eg, specific tension [τ]), the constant for a gracilis-powered elbow flexion is

$$\alpha_{GR} = \alpha_{KE} \times \tau_{GR}/\tau_{KE} \quad (3)$$

where α_{GR} is the constant for the gracilis muscle, τ_{GR} is the specific tension for the gracilis muscle, and τ_{KE} is the specific tension for the quadriceps muscle group. Using the average values of specific tension as published in Persad et al's systematic review[24] for the quadriceps muscle group, and the gracilis, the constant for the gracilis muscle is

$$\alpha_{GR} = 0.41$$

Therefore, equation 2 for a gracilis-powered elbow flexion can be expressed as

$$f_{MVC_{GR}} = 244.57 \times EMG_{norm_{GR}} \quad (4)$$

where $f_{MVC_{GR}}$ is the force generated by the gracilis muscle as a percentage of the MVC, and $EMG_{norm_{GR}}$ is the normalized muscle activity in the gracilis muscle. Therefore, MET_{avg} for a gracilis-powered elbow flexion ($MET_{avg_{GR}}$) can be expressed as (combining equations 1 and 4),

$$MET_{avg_{GR}} = 0.11 \times EMG_{norm_{GR}}^{-1.601} \quad (5)$$

The $MET_{avg_{GR}}$ was calculated for all the participants using the $EMG_{norm_{GR}}$ for isometric flexion of the elbow. The $MET_{avg_{GR}}$ for trials where the participants could not perform the task was recorded as zero minutes. The difference between the $MET_{avg_{GR}}$ with and without PMEO was calculated as

$$\Delta MET_{avg_{GR}} = MET_{avg_{GR}}(w/PMEO) - MET_{avg_{GR}}(w/o\ PMEO)$$

Outcome measures

Fatigue was assessed in the transferred muscle using a combination of spectro-temporal and duration-based measures. In alignment with established literature, both the MPF_{GR} and $iEMG_{GR}$ of the gracilis were monitored, as a decrease in MPF_{GR} and an increase in $iEMG_{GR}$ are characteristic indicators of muscle fatigue.[10-12] In addition, the $MET_{avg_{GR}}$ was used as a surrogate metric for fatigue, as it estimates the maximum duration the gracilis can sustain a specific force output.

Statistical analysis

Task success rates with and without the PMEO were analyzed to assess the association between the use of the PMEO and task completion. A 2×2 contingency table was constructed with the number of successful versus unsuccessful trials as columns and the orthosis condition (with PMEO vs without PMEO) as rows. A right-tailed Fisher's exact test was used to determine if the odds of task success were higher with the use of the PMEO.

All data were processed in MATLAB 2021b (The MathWorks, Inc.).[f]

Results

Participants and successful trials

Eight participants fulfilled the inclusion/exclusion criteria, with 1 participant excluded because of very weak elbow flexor activity, precluding the operation of the PMEO. Hence, the data for 7 participants were analyzed (table 1).

Fisher's exact test revealed that participants were significantly more likely to complete the task successfully when using the PMEO, with an odds ratio of 6.91 (*P*=.005). The participants successfully completed a median of 3 trials while using the PMEO, lifting an average weight of 2.1±1.8 kg. In contrast, without the PMEO, they completed only 1 trial on average, lifting 1.1±1.9 kg (table 2).[4] Representative participant data showed that the brake mechanism allowed the participants to relax their elbow flexor while the PMEO held the elbow at the set angle (fig 3). These results suggest that the powered orthosis significantly improved the participants' ability to sustain repeated lifting tasks. Additional kinematics and range of motion data have been presented in a companion article.[4]

EMG's spectro-temporal metrics

The spectro-temporal EMG properties of the gracilis muscle (MPF_{GR} and $iEMG_{GR}$) suggested reduced fatigue during sustained elbow flexion when assisted by the PMEO. Median power frequency (MPF_{GR}) was higher with the PMEO in 4 of the 7 participants (3 more impaired and 1 less-impaired), indicating a delayed onset of fatigue (fig 4A). Similarly, integrated EMG ($iEMG_{GR}$) was lower with the PMEO in the same 4 participants (fig 4B).

Table 1 Participant demographics.

	More Impaired	Less Impaired
N	4	3
Height (m)	1.8±0.0	1.8±0.0
Weight (kg)	91.8±5.1	107.0±13.6
Age (y)	38±6	29±5
Time since surgery (mo)	38±8	49±13
Affected side (right/left)	0/4	0/3
Type of injury (N)	UT: 1; PP: 3	UT: 0; PP: 3
Nerve connected to the elbow flexor (N)	SAN: 3; ICN: 1	SAN: 1; ICN: 2
PMEO weight on arm (kg)	0.55±0.03	0.55±0.02
Weight lifted with PMEO (kg)	1.3±0.3	3.1±1.2
Weight lifted without PMEO (kg)	0.2±0.1	2.4±1.2

Abbreviations: ICN, intercostal nerve; PP, pan plexus (Injury to C5-C8 and T1); SAN, spinal accessory nerve; UT, upper trunk (injury to C5-C7).

Table 2 Trials and performance per participant (✓=success; ✘=failure).

		With PMEO					Without PMEO				
Participant ID	Group	Trial 1	Trial 2	Trial 3	Successful Trials	Average Weight Lifted (kg)	Trial 1	Trial 2	Trial 3	Successful Trials	Average Weight Lifted (kg)
01	LI	✓	✓	✓	3	5.6	✓	✘	✘	1	5.6
02	MI	✓	✓	✓	3	2.2	✘	✘	✘	0	0.0
09	MI	✓	✓	✘	2	0.7	✘	✘	✘	0	0.0
11	LI	✘	✘	✘	0	0.0	✓	✘	✘	1	0.0
12	LI	✓	✓	✓	3	3.6	✓	✓	✓	3	1.7
13	MI	✓	✓	✓	3	1.7	✓	✓	✓	3	0.7
15	MI	✓	✓	✓	3	0.7	✘	✘	✘	0	0.0

Abbreviations: LI, less impaired; MI, more impaired.

Maximum endurance time

$MET_{avg_{GR}}$ during isometric elbow flexion improved with the use of the PMEO, particularly among participants in the more impaired group (fig 5). In this group, the median $MET_{avg_{GR}}$ was 4 minutes (range: 0.3-10.9min) with the PMEO and 0 minutes (range, 0-37.3min) without the PMEO (ie, >50% of the subjects had a $MET_{avg_{GR}}$ of 0). For the less impaired group, the median isometric $MET_{avg_{GR}}$ was 2 minutes (range, 0-33.2min) with the PMEO and 2.5 minutes (range, 0.1-14.3min) without the PMEO (fig 5A). The median $\Delta MET_{avg_{GR}}$ was 3.8 minutes (range, −36.7 to 10.9min) for the more impaired group and −0.13 minutes (range, −0.6 to 19min) for the less impaired group (fig 5B).

Discussion

This study demonstrated that the PMEO effectively improved elbow flexion fatigue resistance in individuals with BPI and subsequent FFMT. Across all tested metrics, consistent results were observed, showing enhanced fatigue resistance in 4 out of 7 participants with the use of the PMEO.

Performance of trials

Three of the more impaired participants were unable to contract their elbow without the assistance of the PMEO. All other participants were able to perform the lifting task at least once. Overall, the PMEO enhanced participants' ability to perform the task, with only 1 individual unable to complete it successfully, even with the device. The brake mechanism in the PMEO likely contributed to this improvement by locking the elbow in place and reducing the muscular effort needed to sustain a flexed position.[4,5] Although participants were expected to maintain the isometric contraction indefinitely with the aid of the brake mechanism, their unfamiliarity with the orthosis and lack of formal training may have limited their performance. In addition, the PMEO did not stabilize the shoulder. Patients with a BPI who used an assistive device for their elbow expressed "shoulder pain after a short period of time," also stating that they experienced a

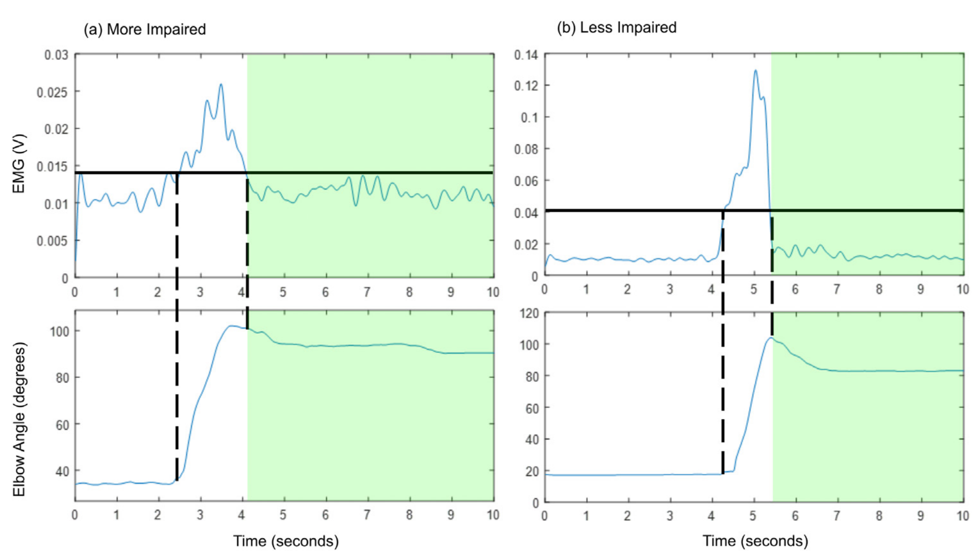

Fig 3 Representative participants from the (A) more impaired and (B) less impaired groups. The black line denotes the EMG activation thresholds for the respective participants. The shaded green area denotes the period of sustained elbow flexion angle with a relaxed gracilis muscle, proving the brake within the PMEO was holding the elbow at the set angle.

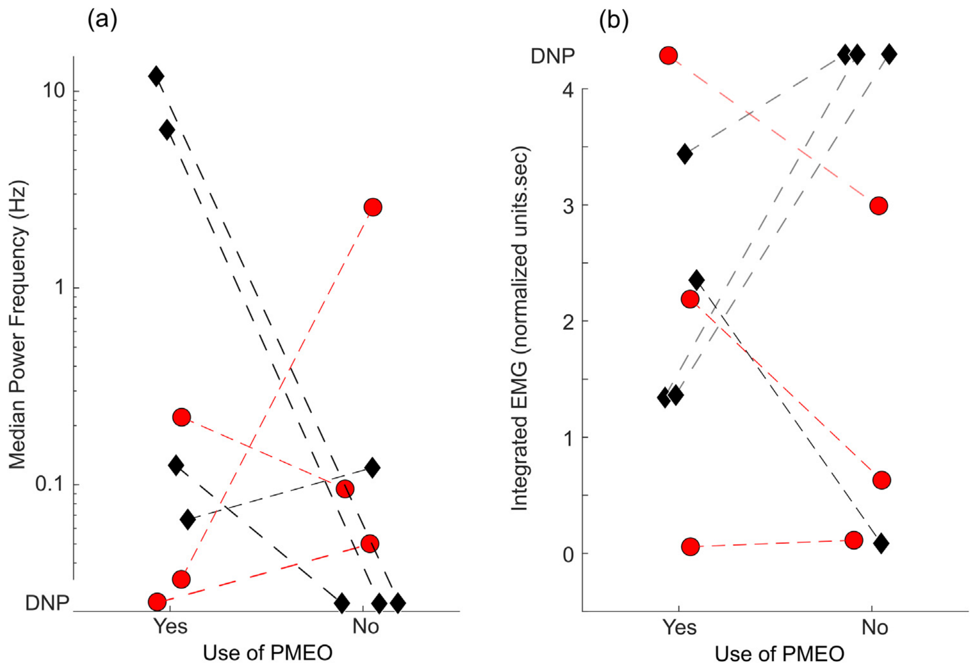

Fig 4 The gracilis elbow flexor's EMG spectral and temporal properties during isometric contraction (A) Median power frequency (MPF_{GR}) in Hz (y-axis in log-scale). The MPF_{GR} should reduce with an increase in fatigue. (B) Integrated EMG $iEMG_{GR}$ in seconds. The $iEMG_{GR}$ should increase with an increase in fatigue. Abbreviation: DNP, did not perform.

nonphysiological load on the shoulder, putting a strain on the joint.[25] These observations point to the potential of additional orthotic training and habituation to improve the PMEO's capacity to aid its user during ADLs.

Change in the EMG's spectro-temporal metrics

In this study, 4 participants demonstrated lower MPF_{GR} and higher $iEMG_{GR}$ when performing the task without the PMEO, indicating a reduced fatigue resistance in the absence of orthotic support. A decline in MPF is a well-established marker of muscle fatigue.[26] MPF was widely used in recent fatigue analysis performed on athletes,[27–29] as well as patients with disabilities.[30,31] Similarly, iEMG has been shown to increase progressively with fatigue, as described by Viitasalo et al.[12] The combination of reduced MPF and elevated iEMG in the participants of this study suggested that the PMEO may have helped delay the onset of fatigue by decreasing the neuromuscular demand on the transferred gracilis muscle. Notably, this study provided one of the first applications of spectro-temporal EMG metrics to evaluate fatigue resistance in a FFMT, filling a gap in the existing literature.

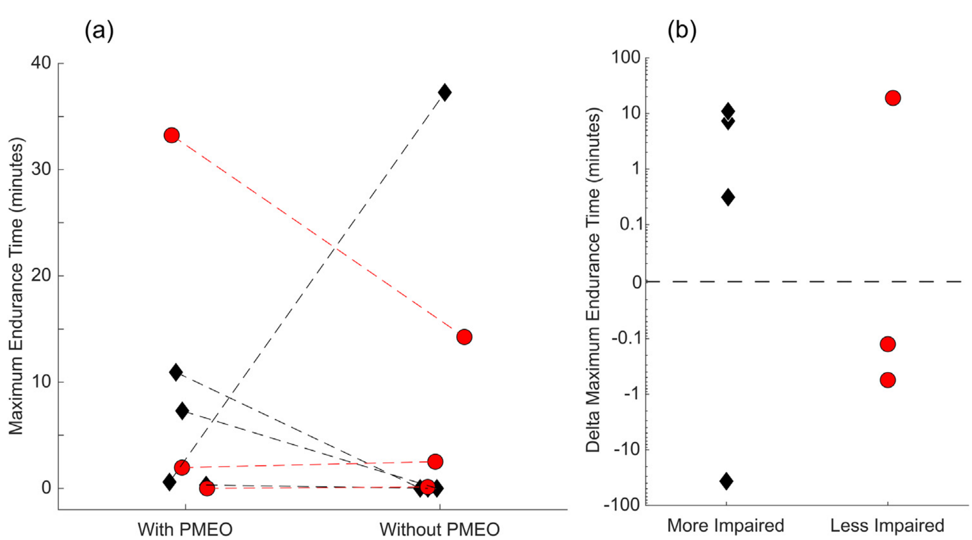

Fig 5 Maximum endurance time for the gracilis elbow flexor during isometric contraction (A) $MET_{avg_{GR}}$ with versus without the PMEO in minutes, (B) $\Delta MET_{avg_{GR}}$ in minutes (y-axis in log-scale).

Improvement in MET from the use of orthosis

The PMEO increased MET in individuals with BPI and FFMT. It was particularly beneficial for participants in the more impaired group, where the median MET increased from 0 minutes without the PMEO to 4 minutes with it. These findings aligned with previous literature demonstrating the efficacy of orthotic devices in enhancing muscular endurance. For example, a spinal orthosis used to manage hyperkyphosis in elderly adults increased endurance time from 7 to 11 seconds during the Lumbar Trunk Muscle Endurance Test developed by Ito et al.[32,33] Similarly, a semirigid backpack-type thoracolumbar orthosis worn for 3 months improved both endurance and muscle strength in adults with hyperkyphosis.[34] In another study, walking endurance improved from 108±18 minutes to 313±32 minutes after 12 weeks of training with an isocentric reciprocating gait orthosis in patients with spinal cord injury.[35] Among individuals with less severe forms of BPI, reconstructed with a nerve transfer for elbow flexion, postsurgical elbow flexion endurance times showed considerable variability, ranging from 5 to 20 minutes.[36]

Limitations

The measures used in this study served as estimations of the gracilis muscle's ability to resist fatigue after FFMT reconstruction for elbow flexion. Although a more extensive and prolonged fatigue assessment may provide more precise insights into the muscle's fatigue properties, the current estimations are supported by prior literature.[6,10-12,23] In addition, individuals with BPI typically exhibit poor fatigue resistance in their muscles.[37] Spectro-temporal metrics were not assessed over time because of the significant alterations in the relationship between surface EMG activity and mechanical force in the elbow flexor after FFMT reconstruction compared with normal biceps.[38,39] Such physiological deviations render the interpretation of traditional time-dependent EMG shifts unreliable within this specific clinical population. The PMEO did not improve fatigue resistance metrics for all participants, likely reflecting the inherent variability in injury severity and recovery among individuals with BPI.[1,2]

Brachial plexus injuries are not common in the general population. The PMEO was evaluated in this patient population because of the impact of enhancing function in these patients. The sample size may seem limited, but it is in line with the broader literature on adult BPI and FFMT interventions, where patient recruitment remains inherently constrained.[38−42] Even classical reports of FFMT for elbow flexion after complete plexus avulsion involved only 27 patients undergoing 36 procedures >10 years.[43] The rarity of sufficiently homogeneous BPI cases, combined with the heterogeneity of injury severity, surgical repair history, and functional residuals, makes larger prospective cohorts difficult to assemble. Given these constraints, our cohort of 7, although modest in size, is meaningful: it is well-characterized (with stratification by impairment via MMT), and it leverages detailed spectro-temporal EMG and energetics in a way rarely practical in larger, more heterogeneous groups. Although we acknowledge that the sample size is limited, our findings can be seen as robust initial evidence for the PMEO's potential to enhance endurance and functional outcomes after FFMT in BPI patients.

Conclusion

This pilot study demonstrated that the PMEO improved elbow flexion fatigue resistance in individuals with BPI and FFMT. The PMEO enabled task performance in more impaired individuals, with increased endurance times and reduced muscular effort because of its brake mechanism. Although limited by a small sample size and lack of shoulder stabilization, these findings align with prior research and highlight the PMEO's potential to enhance functional endurance in this population.

Suppliers

a. EMG500; Motion Lab Systems, Inc.
b. Tegaderm 1624W; 3M.
c. Tape Underwrap SKU: 214546; Cramer Sport Medicine.
d. Raptor-12; Motion Analysis Corporation.
e. Visual 3D; C-Motion, Inc.
f. MATLAB 2021b; MathWorks, Inc.

Keywords

Elbow flexion assistance; Fatigue resistance; Muscle transfer; Myoelectric elbow orthosis; Rehabilitation

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Authorship contributions/CRediT statements

S.G.B. collected the data, conducted part of the analysis, and drafted the initial manuscript. D.N. contributed to data analysis and writing sections of the manuscript. A.Y.S. and K.R.K. supervised the project and provided critical feedback. All authors participated in the writing and review process.

Data statements

Data published in this article will be made available upon reasonable request to the corresponding author.

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